

A Survey on Document Image Binarization Techniques

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Abstract—Document image binarization is performed to segment foreground text from background text in badly degraded documents. In this paper, a comprehensive survey has been conducted on some state-of-the-art document image binarization techniques. After describing these document images binarization techniques, their performance have been compared with the help of various evaluation performance metrics which are widely used for document image analysis and recognition. On the basis of this comparison, it has been found out that the adaptive contrast method is the best performing method. Accordingly, the partial results that we have obtained for the adaptive contrast method have been stated and also the mathematical model and block diagram of the adaptive contrast method has been described in detail.

Keywords—segmentation, document image binarization, degraded document image, image contrast

I. INTRODUCTION

Document image binarization is performed in the preprocessing stage for document analysis.[2] It is used to perform optical character recognition. The main objective of any document image binarization technique is to segment the foreground text from the background text. The thresholding of degraded documents is a major challenge. Thresholding refers to the conversion of gray-scale image to a binary image. Several historical documents suffer from degradation as they begin to age over a period of time. Document images suffer from uneven illumination, image contrast variation, bleed-through and smear. Two of these effects have been illustrated in Fig 1.and 2 below

Fig. 1.Documents suffering from bleed through

As illustrated in Fig.1 the text suffers from bleed through where the ink from the other side seeps through the front.

In Fig. 2 the document image is degraded by smear making the text in it almost indecipherable

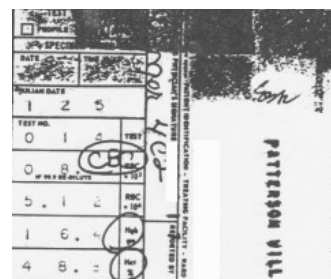


Fig 2.Document suffering from smear

Thus, these different types of document degradations result in document thresholding error and makes degraded document image binarization a formidable challenge for many state of the art techniques.

The rest of the paper is organized as follows. Current state of the art binarization techniques have been reviewed in section II.Evaluation results obtained on dataset of DIBCO 2011 have been reported in section III and section IV. Section V describes the mathematical model for the adaptive contrast method in detail which is followed by the block diagram of adaptive contrast method in section VI and partial results obtained by the authors for this method in section VII.

II. REVIEW OF BINARIZATION TECHNIQUES

A Otsu's Method N. Otsu, (1979) [8]

In Otsu's method cluster-based image thresholding, has been used for the reduction of a gray level image to a binary image. This algorithm tries to reduce combined spread (intra class variance) by assuming that the image contains two classes of pixels. It assumes that an image follows a bimodal histogram i.e. it contains foreground and background pixels. It then calculates the optimum threshold separating the two classes to ensure that its combined spread is minimal. This method gives acceptable results when the pixels in each class are close to each other. The limitations of this method are that many degraded documents do not have a clear bimodal pattern. Also another limitation is that minimization of intra class variances maximizes between class scatter. [3]

B.Niblack's Method W.Niblack (1986) [4]

By sliding a rectangular window over the gray level image Niblack's algorithm calculates a pixel wise threshold.

The threshold T is computed by using the mean m and standard deviation s , for all the pixels within the window, and this threshold is denoted as:

$$T = m + k \times s$$

where k is a constant, which determines how much of the total print object edge is retained, and has a value between 0 and 1. The value of k and the size SW of the sliding window defines the quality of binarization [9]. The limitation of Niblack's method is that the resulting binary image suffers from a great amount of background noise especially in areas without text. [10]

C.Sauvola's Method J. Sauvola and M. Pietikainen (2000) [5]

In this method the page is considered as a collection of subcomponents such as text, background and picture. To define a threshold for each pixel of the background and pictures a soft decision method is used. To define a threshold for each pixel of textual and line drawing areas a text binarization method is used. Finally the results of these algorithms are combined. [5]. Although this method solves the problem posed by Niblack's approach but in many cases the characters become extremely thinned and broken. [10]

D.Bernsen's Method J. Bernsen (1986)

In this method the local image contrast is defined as follows

$$C(i, j) = I_{\max}(i, j) - I_{\min}(i, j)$$

where $C(i, j)$ denotes the contrast of an image pixel (i, j) .

$I_{\max}(i, j)$ and $I_{\min}(i, j)$ denote the maximum and minimum intensities within a local neighborhood window of (i, j) respectively. The pixel will be classified into text or background by comparing with the mean of $I_{\max}(i, j)$ and $I_{\min}(i, j)$. If the local contrast $C(i, j)$ is smaller than the threshold then the pixel is set as background and vice-versa. This method is simple. But the limitation of this method is that it does not work properly on degraded document images with a complex background. [2]

E.Local Minimum Maximum method (LMM) Method S. Lu, B. Su, and C. L. Tan (2010) [7]

This method is an improvement over Bernsen's method and handles the documents with a complex background well. In this method the local image contrast introduces a normalization factor. This normalization factor compensates for the image variation within the document background. Here the local image contrast is evaluated as follows:-

$$C(i, j) = \frac{I_{\max}(i, j) - I_{\min}(i, j)}{I_{\max}(i, j) + I_{\min}(i, j) + \varepsilon} \quad (1)$$

where ε is a positive but infinitely small number that is added in case the local maximum is equal to 0. [2] In particular, the numerator (i.e. the difference between the local maximum and the local minimum) captures the local image difference that is

similar to the traditional image gradient. The denominator acts as a normalization factor that lowers the effect of the image contrast and brightness variation. For image pixels within bright regions around the text stroke boundary, the denominator is large, which neutralizes the large numerator and accordingly results in a relatively low image contrast. But for image pixels within dark regions around the text stroke boundary, the denominator is small, which compensates the small numerator and accordingly results in a relatively high image contrast. [7]

As a result, the contrasts of image pixels (lying around the text stroke boundary) within both bright and dark document regions converge close to each other and this facilitates the detection of high contrast image pixels lying around the text stroke boundary. [7]

The limitation of this method is that, it cannot handle document images with bright text having bright background properly. A weak contrast is calculated for image pixels having bright text stroke edges and which lie within bright regions. A large denominator and a small numerator are produced for documents having bright text stroke edges and which lie within bright regions. This problem is known as over-normalization problem.

F. Adaptive Contrast (AC) Method Bolan Su, Shijian Lu, and Chew Lim Tan (2013) [2]

This method overcomes the over-normalization problem posed by LMM method. This method is comprised of four steps. In the first, step an adaptive image contrast which is a combination of the local image contrast and the local image gradient is constructed for an input degraded document image. In the second step, the contrast map is then binarized using Otsu's method and then it is combined with Canny's edge map to identify the text stroke edge pixels. In the third step, the document text is further segmented by a local threshold that is estimated based on the intensities of detected text stroke edge pixels within a local window. In the final step some post processing is applied to produce better results. This method was one of the best performers in ICDAR 2011 binarization contest. [2]

III. PERFORMANCE METRICS

In this section, we have compared the performance of document image binarization techniques discussed in section II with the help of various performance metrics. The ensembles of metrics which are suitable for above mentioned purpose are (i) F-Measure (ii) PSNR (iii) Distance Reciprocal Distortion Metric (iv) Misclassification Penalty Metric. These metrics are described below:

G. F-Measure

$$F\text{-Measure} = \frac{2 \times \text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}}$$

$$\text{Where Recall} = \frac{TP}{TP + FP}, \text{ Precision} = \frac{TP}{TP + FN} \quad (4)$$

TP, FP, FN denote the True positive, False positive

and False Negative values, respectively. [1]

H. PSNR

$$\text{PSNR} = 10 \log \frac{C^2}{\text{MSE}} \quad (5)$$

$$\text{Where MSE} = \frac{\sum_{x=1}^M \sum_{y=1}^N (I(x, y) - I'(x, y))^2}{\text{MN}}$$

PSNR is a measure of how close is an image to another. Therefore, the higher the value of PSNR, the higher is the similarity between the two images being compared. C is the difference between foreground and background [1]

I. Distance Reciprocal Distortion Metric (DRD)

The Distance Reciprocal Distortion Metric (DRD) has been used before to measure the visual distortion in binary document images [1]. It properly correlates with the human visual perception and it measures the distortion for all the S flipped pixels as follows:

$$\text{DRD} = \frac{\sum_{k=1}^S \text{DRD}_k}{\text{NUBN}} \quad (6)$$

where DRD_k is the distortion of the k^{th} flipped pixel and it is calculated using a 5x5 normalized weight matrix W_{Nm} as defined in [13]. DRD_k equals to the weighted sum of the pixels in the 5x5 block of the Ground Truth GT that differ from the centered k^{th} flipped pixel at (x,y) in the Binarization result image B .

$$\text{DRD}_k = \sum_{i=-2}^2 \sum_{j=-2}^2 |GT_k(i, j) - B_k(x, y)| \times W_{Nm}(i, j) \quad (7)$$

Finally, NUBN is the number of the non-uniform (not all black or white pixels) 8x8 blocks in the GT image.[1]

J. Misclassification penalty metric (MPM)

The Misclassification penalty metric MPM evaluates the prediction against the Ground Truth (GT) on an object-by-object basis. Misclassification pixels are penalized by their distance from the ground truth object's border

$$\text{MPM} = \frac{\text{MP}_{\text{FN}} + \text{MP}_{\text{FP}}}{2} \quad (8)$$

$$\text{where } \text{MP}_{\text{FN}} = \sum_{i=1}^{\text{NFN}} d_{\text{FN}}^i / D, \text{MP}_{\text{FP}} = d_{\text{FP}}^i / D$$

d_{FN}^i and d_{FP}^i denote the distance of the i^{th} false negative and the j^{th} false positive pixel from the contour of the GT segmentation. The normalization factor D is the sum over all the pixel-to-contour distances of the GT object. A low MPM score denotes that the algorithm is good at identifying an object's boundary.[1]

IV. RESULTS OBTAINED AFTER TESTING ON DATASET OF DIBCO 2011

In this section the results obtained after testing on dataset of DIBCO 2011 have been stated. The results below are cited from the Document Image Binarization Contest DIBCO 2011 held under the framework of the International Conference on Document Analysis and Recognition 2011.[1]

TABLE I
EVALUATION RESULTS OF THE DATASET FOR DIBCO 2011 [2]

Methods	F-Measure %	PSNR	DRD	MPM
Otsu [8]	82.22	15.77	8.72	15.64
Sauvola [5]	82.54	15.78	8.09	9.20
Niblack [4]	68.52	12.76	28.31	26.38
Bernsen [6]	47.28	7.92	82.28	136.54
LMM[7]	81.67	15.59	11.24	11.40
AC [2]	87.8	17.56	4.84	5.17

On the basis of the comparison we have found out that the adaptive contrast method is the best performing method. Hence, in the subsequent sections of this paper we have explored this method in detail.

V. MATHEMATICAL MODEL FOR ADAPTIVE CONTRAST METHOD.

This section describes the mathematical model for the adaptive contrast method. As already discussed in section II the LMM method suffers from over-normalization problem. To surmount this over-normalization problem posed by LMM method an adaptive local image contrast [1] is calculated. This adaptive local image contrast is a combination of local image contrast and local image gradient. The adaptive image contrast is calculated as follows:-

$$C_a(i, j) = \alpha C(i, j) + (1 - \alpha)(I_{\text{max}}(i, j) - I_{\text{min}}(i, j)) \quad (9)$$

where $C(i, j)$ denotes the local contrast and $(I_{\text{max}}(i, j) - I_{\text{min}}(i, j))$ refers to the local image gradient that is normalized to [0, 1]. [1]. The value of α is based on the document image's statistical information and it is the weight between local contrast and local gradient. If the document has

a high has a considerable amount of intensity variation in it the image contrast is assigned a high weight. Thus this technique will rely more on local image contrast and it will be able to capture the intensity well. On the contrary, if the document does not have a significant intensity variation then the image gradient will be assigned a higher weight. The mapping from document image intensity variation to α by a power function is defined as follows:

$$\alpha = \left(\frac{Std}{128}\right)^{\gamma} \quad (10)$$

where Std denotes the document image intensity standard deviation, and γ is a pre-defined parameter. [2]
The value of γ can lie in the interval $[0, \infty]$. Local image contrast will play a major role when γ has a small value. The local image gradient will play a major role when γ has a large value.

VI. BLOCK DIAGRAM OF ADAPTIVE CONTRAST METHOD

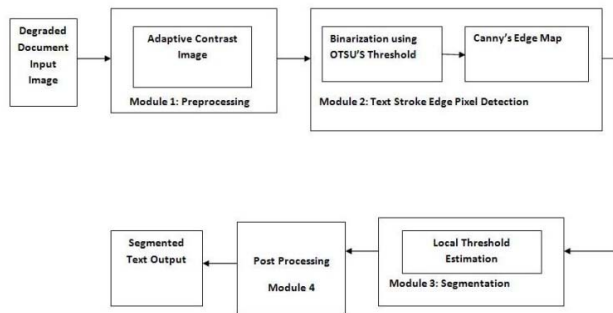


Fig 3. Block diagram of adaptive contrast method.

In this section we have described the block diagram of the adaptive contrast method. The block diagram of adaptive contrast method is as shown in Fig.3. The block diagram is composed of four modules namely preprocessing, text stroke edge pixel detection, segmentation and post processing modules respectively.

The input is a degraded document image. In the preprocessing module an adaptive image contrast is calculated using equations (9) and (10). The constructed contrast image has a clear bimodal pattern.[2]. Hence in the text stroke edge detection module the text stroke pixel candidate is detected by using Otsu's global thresholding method. This binarized map obtained after using Otsu's global thresholding method is further enhanced by using Canny's edge detector which matches edges close to real edges in the detecting image. Then in the segmentation module an algorithm for local threshold estimation is used. In this algorithm the most frequent distance between two adjacent pixels in horizontal direction is calculated and then the stroke width is estimated.

In the post processing module the isolated foreground pixels that do not connect with other foreground pixels are filtered

out to make the pixel edge set accurate. Also, the neighborhood pixel pair that lies on symmetric sides of a text stroke edge pixel are made to belong to different classes (i.e., either to the document background or the foreground.) One pixel of the pixel pair is therefore labeled either as foreground or background if both the pixels belong to the same class.[1]

VII. RESULTS AND DISCUSSION

In this section the partial results that we have obtained for Module 1 & partly for Module 2 have been shown.

Fig.4 shows the degraded input image.

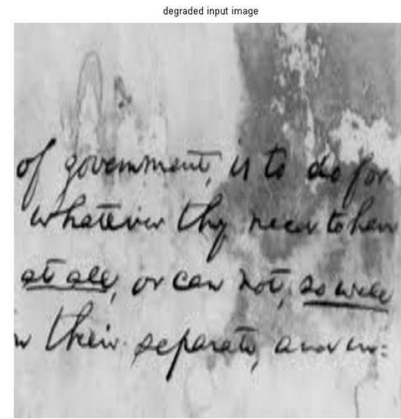


Fig 4. Degraded input image

Fig.5 shows the contrast image that has been obtained.

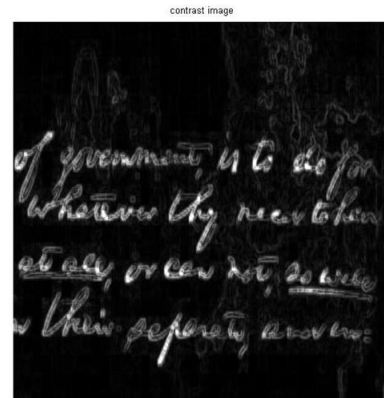


Fig.5 Contrast Image

Fig.6 shows the gradient image that has been obtained



Fig.6 Gradient image

Fig.7 shows the adaptive image contrast that has been obtained



Fig.7 Adaptive Image Contrast

Fig.8 shows the result that has been obtained after applying Otsu's global thresholding method.

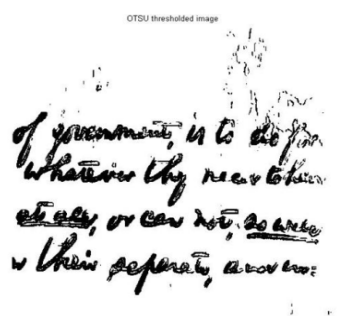


Fig.8Otsu's thresholded image

VIII. CONCLUSION

Thus ,this paper presents a survey of some of the widely used document image binarization techniques. It compares the various existing techniques with this method using various evaluation measures such as F-Measure, PSNR, DRD, MPM.The inference that can be drawn from this comparison is that all the methods listed above have some advantages and some drawbacks. The adaptive contrast method outperforms most of the document image binarization methods in terms of F-measure, PSNR, DRD and MPM. Hence,we have described this method in detail. In our future work we will improve the quality assessment for the extracted document .

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